



Shakil Ahmed • Principal Systems Engineer

Former Engineering Team Lead at Legiit (scaled backend architecture to \$1M ARR across 1,500+ businesses & 1,000,000+ transactions). Securiti Certified AI Architect.

HOURLY RATE

\$40.00 / hr

Client Avg: \$50.00/hr

WEEKLY AVAILABILITY

20–25 hrs / wk

EST / EST-Friendly

OPERATING MODEL

Zero Micromgmt

Async Standups & CI

PHASE 1 EXECUTION: 30-DAY SYSTEMS RELIABILITY & PERFORMANCE ROADMAP

Zero Disruption to Active Users

TIMELINE	STRATEGIC OBJECTIVE	CONCRETE ENGINEERING DELIVERABLES	TARGET OUTCOME
Week 1	Codebase Discovery & Observability Baseline	Full repository audit (FastAPI/Node), environment inventory, local Docker Compose setup, OpenTelemetry tracing, slow query logging (>200ms), and GitHub Actions CI workflow enforcement.	100% CI automated; P99 baseline established
Week 2	PostgreSQL 16 Profiling & Index Optimization	Run EXPLAIN (ANALYZE, BUFFERS) on top 5 slowest queries. Replace sequential table scans with composite covering B-Trees. Deploy via CREATE INDEX CONCURRENTLY with zero table locking.	112x speedup (428ms → 3.8ms); 0 lockouts
Week 3	Asynchronous Queues & Ingestion Decoupling	Decouple high-throughput webhooks and external API sync into Redis Streams & Celery/Inngest workers. Implement exponential backoff retries with jitter and Dead-Letter Queue (DLQ) alerts.	5,000 req/s surge; 0 dropped payloads
Week 4	Connection Pooling, Caching & AI Microservices	Deploy PgBouncer transaction pooling (multiplexing 1,450 web connections to 25 Postgres slots). Implement XFetch probabilistic early expiration in Redis. Deploy multi-provider AI circuit breaker.	0 connection starvation; <500ms AI failover

ARCHITECTURE & RESILIENCE STANDARDS IMPLEMENTED IN LIVE COCKPIT

Live Verification: scaleops-core.vercel.app

PostgreSQL 16 Buffer Optimization

Eliminates runaway AWS RDS IOPS costs by converting 14,200 disk buffer reads into 42 RAM buffer cache hits through selective partial and composite B-Tree indexes.

P99 Query: 3.8ms • Zero Table Locks

PgBouncer Connection Multiplexing

Prevents FATAL: too many connections under high concurrency by recycling lightweight transaction connections into a pooled 25-slot database backend.

58:1 Ratio • 14MB RAM Overhead

Redis XFetch Stampede Shield

Replaces dangerous hard-TTL key expiration with probabilistic early recomputation, ensuring exactly 1 worker refreshes hot keys while all clients read warm cache.

0 DB Spike Events • 92% Cache Hit Ratio

STAFF AUGMENTATION OPERATING CONTRACT & AUTONOMOUS STANDARDS

Upwork Hourly Contract Structure

ZERO MICROMANAGEMENT EXECUTION Self-directed backlog prioritization, atomic Git branches, strict typing, passing lint/test suites, and detailed daily async standup summaries posted to Slack/ClickUp.	COMMUNICATION & SYNC CADENCE Overlapping core hours in North American timezone (EST). 4-hour PR review SLA. Available for weekly sprint demos or technical architectural syncs as needed.	TRANSPARENT HOURLY BILLING Tracked via Upwork Desktop App (or manual time if authorized). Detailed work notes per memo. Calibrated at \$40.00/hr (aligned with client's \$50/hr historical average).
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